Auditing Recovery after Depth Pruning: Perplexity, Evaluation Interfaces, and Construction Matter

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Abstract

Perplexity and downstream performance can diverge after compression, but post-pruning recovery is still commonly summarized by a loss curve and a small set of endpoints. We present a *multi-interface recovery audit* that jointly follows same-source perplexity, two MMLU scoring interfaces, and zero-shot closed-book QA while pairing them with an intact continued-pretraining control, a random null floor, and an alternative same-depth construction. On OLMo-2-7B, a 16-layer keep14+fresh2 model improves through 200k steps to PPL 10.561, yet reaches only .319 MMLU versus .605 for the intact base and remains 11.6–34.2 points behind on PopQA, TriviaQA, and NQ-open. Complete-option scoring raises keep14 by 6.48 points, but raises a fully random 16-layer model by 11.28 points to the same .360 content-score floor, showing why an interface gain needs a null baseline. At the same nominal depth and displayed 200k-step budget, ShortGPT reaches PPL 9.780 and is 15.5 points stronger on letter MMLU but only 1.8 points stronger on content MMLU than keep14; this contrast jointly changes inherited-block count, non-contiguous selection, final-block retention, and fresh-tail use. It also improves all three closed-book tasks but remains far below the intact references. The likelihood ordering transfers to WikiText-103 and PG-19, while a 15.505B-token continuation-stream audit finds too little benchmark overlap to explain the gaps. The intact full32 branch remains much closer to the base at its reported 200k endpoint. Together, these findings establish a practical measurement lesson: recovery should be audited across likelihood, target behavior, evaluation interface, construction, budget, and run-level uncertainty rather than certified by perplexity improvement alone.

1 Introduction

Removing transformer blocks and then continuing pretraining (CPT, often called "healing") is an attractive route to smaller language models (Gromov et al., 2025; Kim et al., 2024; Men et al., 2025; Yang et al., 2024). Held-out perplexity is a natural training signal, but using its improvement as evidence of target recovery makes a stronger measurement assumption: that likelihood and the target evaluation recover together after the structural intervention.

Compression research has already established that perplexity can miss changes in parametric knowledge, downstream behavior, and safety (Namburi et al., 2023; Jaiswal et al., 2024; Xu et al., 2024). Depth-pruning work also reports recovery curves and loss–task gaps (Gromov et al., 2025; Kim et al., 2024; Sreenivas et al., 2024; Wibowo et al., 2025). The open measurement problem is therefore not whether the metrics can disagree, but *how to audit a recovery claim*: is the apparent gap a property of the training path, the evaluation interface, or the exact pruned construction?

We answer this question with a densely instrumented OLMo-2-1124-7B case study (OLMo Team, 2025). Among the post-depth-pruning studies we identified, none jointly combines (i) same-source held-out perplexity and within-path checkpoints, (ii) answer-letter and complete-option MMLU on the same 14,042 items, (iii) three no-retrieval closed-book QA datasets, and (iv) an intact CPT branch, same-shape null points, and an alternative 16-layer construction. This combination turns a familiar "PPL versus task" comparison into a multi-axis test of proxy, interface, and construction.

Figure 1 previews three measurement findings. First, keep14 continues to improve late in training without approaching the intact target scores. Second, complete-option MMLU makes the compressed models look stronger, but the same interface also gives a random model a high floor. Third, at the same nominal depth and displayed steps, ShortGPT exceeds keep14 by 15.5 points on let-

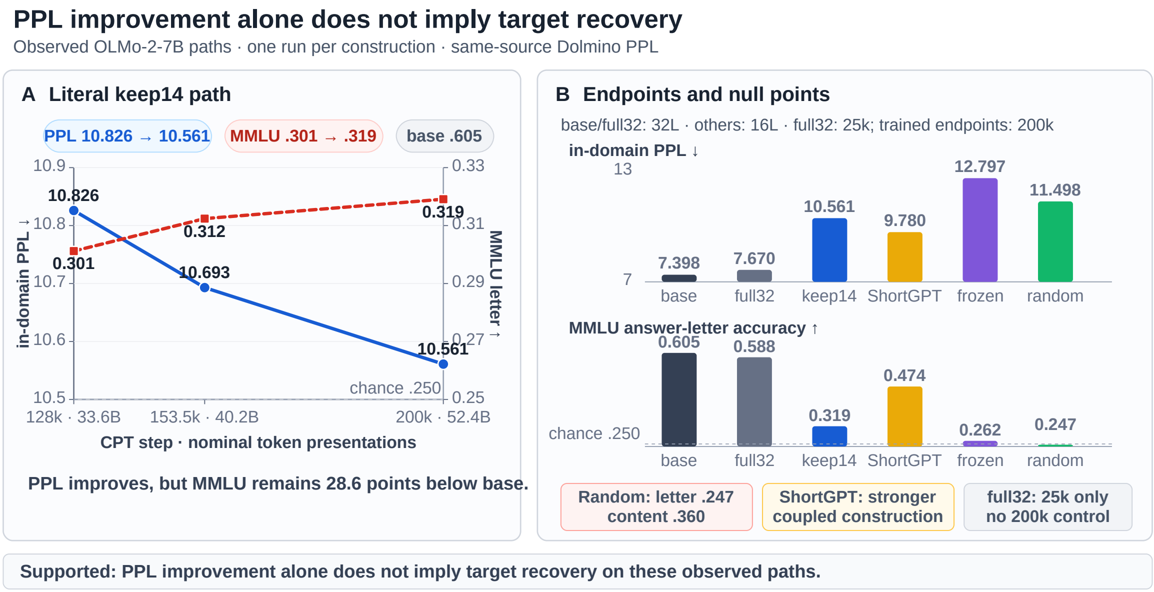

Figure 1: **A multi-axis audit changes the recovery verdict. Left:** keep14 improves from 128k to 200k, yet the large intact-base MMLU gap remains. **Right:** the random model exposes a high complete-option floor, whereas ShortGPT reaches a much stronger endpoint at the same nominal depth and displayed 200k-step budget; full32 supplies the reported 200k intact-CPT reference. PPL uses a same-source held-out Dolmino shard. Each trained construction is one run and token counts are nominal presentations. The exact unmatched factors are reported in Table 2 and Limitations.

ter MMLU but only 1.8 points on content MMLU while changing four construction factors together. Construction and interface are therefore both essential parts of the recovery result.

Study design. The principal construction retains the first 14 of 32 pretrained blocks, appends two fresh blocks, and trains for 200k steps. We compare it with the intact base, a full32 CPT endpoint at 200k, frozen and random same-shape operating points, and a non-contiguous ShortGPT-16 model at 200k. All PPL is in-domain on a disjoint shard from the same Dolmino/DCLM source. Each trained construction is one run; full32 is reported at 200k, and ShortGPT changes four structural choices together. We state these boundaries once here and concentrate the remaining qualifications in Limitations.

Contributions.

- **A multi-interface recovery audit.** We jointly track same-source and out-of-domain likelihood, paired letter/content MMLU, three closed-book QA datasets, and continuation-stream overlap, with item-level uncertainty for the load-bearing MMLU comparisons.
- **Null and construction diagnostics.** An intact CPT reference, random and frozen operating points, and ShortGPT separate three questions that a single loss curve conflates: training-domain fit, interface validity, and construction dependence.
- **Four actionable findings and an audit protocol.** We quantify the late keep14 target gap, the random content-score floor, the ShortGPT–keep14 gaps of 15.5 letter points and 1.8 content points at the same nominal depth/displayed steps, its consistent closed-book advantage, and the much smaller full32 endpoint deficits; we turn them into a six-axis reporting checklist for future recovery studies.

2 Related Work

Table 1 positions our measurement design against the nearest recovery studies. The novelty is the *joint audit*, not the observation that perplexity and downstream metrics can differ.

Selecting and removing depth. ShortGPT and SLEB rank and remove redundant blocks (Men et al., 2025; Song et al., 2024); LaCo merges adjacent layers (Yang et al., 2024); Block-Pruner searches at attention/MLP-block granularity (Zhong et al., 2025). Other studies show that

Study	Intervention	Adaptation	Trajectory	Loss+task	Scratch/init	Intact CPT	MMLU iface.	Closed-book	Construction
Gromov et al.	deep-block removal	CPT	P	Y	N	N	N	N	depth/budget
Shortened LLaMA	depth pruning	CPT/LoRA	Y	Y	Y	N	N	N	ratio/method
Minitron	structured pruning	distillation/CPT	Y	Y	Y	N	N	N	iterative/shape
IteRABRe	iterative block removal	iterative recovery	Y	Y	N	N	N	N	iteration
PASER	pruned-model recovery	selected CPT data	P	Y	N	N	N	N	data policy
This study	prefix+fresh tail	CPT	Y	Y	O	P	Y	Y	coupled 16L

Table 1: **Nearest-work positioning.** Prior work establishes recovery curves, loss–task gaps, initialization effects, and iterative retraining. Among the post-depth-pruning studies we identified, none jointly evaluates a post-depth-pruning path with an intact same-corpus CPT reference, a null operating point, paired MMLU interfaces, closed-book QA, and an alternative same-depth construction. Criteria are binary at the study level: trajectory = at least two post-intervention adaptation checkpoints; loss+task = likelihood and at least one downstream task at a common checkpoint; scratch/init = an explicit initialization comparator; intact CPT = same-corpus continued intact model; MMLU interface = more than one MMLU answer-scoring interface; closed-book = no-retrieval open generation; construction = an explicit comparison among retained structures. Y/N denote criterion present/absent; P denotes only a partial or short-horizon instance; O denotes a confounded operating point rather than a one-factor initialization ablation.

preferred removals depend on task and calibration choices (Siddiqui et al., 2024; Lu et al., 2024; Kim et al., 2026). These works optimize *which* structure to retain. We instead measure what different retained structures recover under CPT.

Recovery paths and retraining controls. Gromov et al. (2025) are the closest antecedent for post-healing loss–task dissociation after removing deeper layers. Shortened LLaMA reports CPT learning curves and compares CPT, LoRA, and pruned versus scratch initialization (Kim et al., 2024). Minitron likewise studies retraining trajectories, iterative pruning, task behavior, and initialization choices within a structured-pruning/distillation pipeline (Sreenivas et al., 2024). IteRABRe explicitly alternates block removal and recovery, plotting task-family trajectories and weak MMLU recovery (Wibowo et al., 2025). Thus neither trajectories nor loss–task dissociation originate here. What these studies do not jointly provide is a same-item interface audit, a random interface floor, closed-book factual generation, an intact same-corpus CPT reference, and a same-depth construction contrast. We assemble those pieces into one auditable recovery study. PASER is complementary: it selects post-training data to improve recovery efficiency rather than auditing the validity of recovery measurements (He et al., 2025). Sheared LLaMA, compact-model distillation, and DRPruning integrate pruning with data allocation, distillation, or robust training (Xia et al., 2024; Muralidharan et al., 2024; Deng et al., 2025). Our prefix-plus-fresh-tail recipe is not presented as competitive with these recovery algorithms.

Repair and concurrent work. LinearPatch and Prune&Comp attribute part of layer-pruning damage to cross-layer interface or magnitude mismatch and repair it with lightweight transformations or compensation (Chen et al., 2025, 2026). A decision- transition analysis studies where post-pruning performance collapse appears (Shi et al., 2026). SlimQwen and ShortOPD appeared within three months of this manuscript version and are treated as concurrent work (Tang et al., 2026; Zhang et al., 2026); they further cover matched-token initialization, progressive recovery, and recognition/generation behavior. ShortOPD and concurrent work on generative reasoning limits strengthen the case for evaluating free generation alongside multiple-choice recognition (Shrestha et al., 2026). Our study complements these recovery methods by asking which measurements justify the word “recovered.”

Evaluating compression beyond perplexity. The Cost of Compression studies how pruning and quantization affect parametric knowledge (Namburi et al., 2023). Jaiswal et al. (2024) show that low perplexity can coexist with large deficits on knowledge-intensive compressed-LM benchmarks, and Xu et al. (2024) document divergent safety effects. These studies motivate multi-dimensional evaluation. Our additional focus is measurement validity during post-depth-pruning CPT: within-arm checkpoints, an available intact same-corpus branch, a random null floor, paired scoring interfaces, closed-book generation, and an alternative same-depth construction. Fragile Knowledge, Robust Instruction-Following reports selective capability effects under width pruning (Martra, 2025); it is adjacent evidence that compression can preserve some interfaces while damaging others. Multiple-choice evaluations are themselves interface-sensitive: first-token letter probabilities can disagree with generated text, leaderboard rankings change with evaluation details, and MMLU varies under answer reordering (Wang et al., 2024; Alzahrani et al., 2024; Gupta et al., 2024). Our content protocol is a diagnostic within this broader literature. Its key addition is the random same-shape floor, which shows that a larger complete-option score is not by itself evidence of inherited target recovery.

3 Observed Constructions and Evaluation

3.1 Model constructions

We study the 32-block OLMo-2-1124-7B base model (OLMo Team, 2025). The principal `keep14+fresh2` construction retains pretrained blocks 0–13, copies the embedding, final normalization, and output head, appends two randomly initialized decoder blocks, and trains all parameters. The resulting model has 16 blocks and 4.060B parameters.

The other rows are *operating points*. full32 inherits all 32 blocks and has no fresh tail. Frozen-front uses the keep14 construction but freezes the 14 inherited decoder blocks; fresh blocks and copied non-block modules remain trainable. Fully random init has the same 16-block shape but inherits no weights. ShortGPT-16 inherits the selected original blocks $[0–12, 16, 17, 31]$, appends no fresh blocks, and trains all parameters. ShortGPT therefore changes inherited count, contiguity/selection, final-block retention, and fresh-tail use together.

3.2 Continued-pretraining paths

Training uses the DCLM portion of `dolmino-mix-1124`, retokenized into 2,048-token windows. A disjoint 4096×2048 shard from the same source is used for in-domain perplexity. Effective batch size is 128; hence 200k steps correspond to 52.4B nominal token presentations. "Nominal" matters because the keep14 resume did not preserve the within-epoch loader offset.

keep14, ShortGPT, frozen-front, random-init, and full32 are reported at 200k. The shallow keep8, keep10, and keep12 arms are also reported at 200k. The 7B pruned endpoints therefore share the same step and nominal-token budget, while remaining unmatched in FLOPs because their depths differ. Table 2 exposes construction, trainable set, learning rate, and budget for every headline row.

Executed keep14 and frozen-front training used peak/minimum LR $2 \times 10^{-5}/2 \times 10^{-6}$ for every trainable parameter because the historical distributed parameter grouping assigned all trainable tensors to the inherited group. Random-init used $10^{-4}/10^{-5}$. ShortGPT and full32 used $2 \times 10^{-5}/2 \times 10^{-6}$. This makes random-init a null operating point rather than an initialization-only ablation. Frozen-front changes the trainable set and is likewise not an adaptation-only ablation.

3.3 Knowledge-sensitive evaluations

Perplexity is raw next-token loss with no chat template or inserted BOS. The same evaluator is run on 289k WikiText-103 and 2.46M PG-19 test tokens. We also scan the exact 15.505B-token Dolmino continuation stream for token 13-/8-gram containment, flagging items at 0.80. Standard MMLU (Hendrycks et al., 2021) displays four lettered options and scores the answer-letter continuation. A content protocol instead uses a label-free prompt and scores each full option text, reporting both summed and per-token mean log-likelihood; the latter is the content headline. The protocols share all 14,042 MMLU test items, but simultaneously change prompt, candidate string, tokenization, and normalization. They are therefore paired interfaces, not a one-factor readout ablation.

PopQA (Mallen et al., 2023), TriviaQA (Joshi et al., 2017), and NQ-open (Kwiatkowski et al., 2019) use zero-shot, no-retrieval generation with the same plain `Question:/Answer:` prompt. PopQA reports normalized-answer containment; TriviaQA and NQ-open report exact match. Exact sample counts, prompts, normalization, software provenance, and unavailable metadata appear in Appendix B.

4 A Multi-Axis Recovery Audit

Let P_t denote in-domain perplexity and E_t a knowledge-sensitive evaluation at checkpoint t. A common informal recovery argument is

$$P_{t_2} < P_{t_1} \quad \Longrightarrow \quad \text{target recovery at } t_2.$$

Here "target recovery" means closing the large deficit to the intact-base score on the same evaluation, not merely obtaining a statistically detectable within-run gain. We audit that argument on three axes: **trajectory** (does the target gap close as PPL falls?), **interface** (does the verdict survive a change in scoring, and what is the null floor?), and **construction** (do models with the same nominal depth reach similar endpoints?). The full32 branch adds an intact same-corpus reference. This design yields diagnostic operating-point comparisons rather than a single scalar recovery score.

5 Results

5.1 Finding 1: likelihood recovery overstates target recovery

Tables 2 and 3 give the central result and its within-path uncertainty. Along the single keep14 path, PPL falls to 10.561 by 200k steps while answer-letter MMLU reaches .319, versus .605 for the intact base. The closed-book endpoints remain lower as well: .142 versus .257 on PopQA, .294 versus .636 on TriviaQA, and .060 versus .205 on NQ-open. These evaluations do not exhaust capability, but jointly show a persistent target gap across one likelihood-based recognition task and three free-generation datasets.

The trajectory result is not "no learning": from 128k to 200k, MMLU gains 1.68 points with a 95% item-bootstrap CI of $[1.08, 2.29]$. The measurement finding is sharper: the statistically detectable gain closes only a small fraction of the 30.41-point deficit present at 128k, leaving 28.62 points at 200k. A loss curve and a significance test therefore answer different questions from recovery to the intact target.

The intact continued-pretraining reference. At the reported 200k endpoint, full32 is 3.7% above the base in PPL and lower by 1.8/0.4 points on letter/content MMLU. Its PopQA, TriviaQA, and NQ-open deficits are 2.9/6.4/4.7 points, versus 11.6/34.2/14.5 points for keep14 at 200k. Thus the intact endpoint is closer to the base than the compressed endpoint. This makes an immediate same-corpus shift an incomplete explanation for the large keep14 gap, while the full32 comparison remains one training run and is not seed-replicated.

5.2 Finding 2: interface gains require a null floor

Table 5 evaluates both MMLU interfaces on exactly the same items. This paired rerun scores keep14 at .3184; the independently merged trajectory/downstream snapshot in Tables 3 and 4 is .3191. We retain both provenance-specific values. Complete-option scoring raises the paired snapshot to .3832, but it raises Random more, from answer-letter chance (.247) to .360. Frozen and Random then become nearly indistinguishable under content scoring despite a statistically significant 1.54-point letter-score gap (Table 8). This reversal changes the interpretation of the interface gain: complete-option likelihood captures useful option-text compatibility, but its high random floor prevents it from certifying inherited target recovery by itself.

The gap recurs under open generation. The base–keep14 deficit is also 11.6 points on PopQA (n=14,267), 34.2 on TriviaQA (n=17,944), and 14.5 on NQ-open (n=3,610), using a shared zero-shot, no-retrieval `Question:/Answer:` protocol. The recurrence across letter scoring, complete-option scoring, and free generation is stronger evidence than any single interface alone.

The likelihood tax also transfers out of domain. Table 6 shows the same model ordering on WikiText-103 and PG-19. For keep14, the PPL ratio to base is $1.43\times$ in-domain, $1.93\times$ on WikiText-103, and $1.52\times$ on PG-19, so in-domain PPL does not hide an OOD reversal. A token 13-/8-gram scan of the exact 15.505B-token continuation stream flags only 1.90/3.45% of MMLU, 0.79/1.38% of TriviaQA, 0.03/0.66% of NQ-open, and 0/0.07% of PopQA at 0.80 containment. Removing flagged MMLU items changes the keep14–Random gap from 7.30 to 7.20/7.13 points. The completed closed-book rescore is equally stable: across all seven evaluated arms, strict-clean TriviaQA/NQ-open scores move by at most 0.15 points and the model ordering is unchanged; PopQA has zero flagged items.

5.3 Finding 3: construction determines the recovery endpoint

What the ShortGPT contrast reveals. At the same nominal 16-layer depth and displayed 200k steps, ShortGPT reaches PPL 9.780 and exceeds keep14 by 15.5 points on letter MMLU but only 1.8 points on content MMLU (Table 7). The contrast jointly changes four factors: inherited-block count, contiguous versus BI-based selection, retention of original block 31, and fresh-tail use. This pattern suggests two testable explanations. First, preserving pretrained blocks—especially the original final block—may retain an interface for answer-letter decisions that two fresh tail blocks must relearn. Second, non-contiguous selection may preserve

Operating point	Construction	Inherited/fresh	Trainable set	Peak LR	Budget	PPL↓	MMLU-L	MMLU-C	PopQA	TriviaQA	NQ-open
Base	intact 32L checkpoint	32/0	none	—	0	7.398	.605	.471	.257	.636	.205
full32	intact CPT	32/0	all	2×10^{-5}	200k / 52.4B	7.670	.588	.466	.228	.572	.158
keep14	prefix14 + fresh2	14/2	all	2×10^{-5}	200k / 52.4B	10.561	.319	.383	.142	.294	.060
ShortGPT	selected $[0\text{–}12, 16, 17, 31]$	16/0	all	2×10^{-5}	200k / 52.4B	9.780	.474	.401	.159	.330	.067
Frozen	prefix14 + fresh2	14/2	fresh2 + non-block	2×10^{-5}	200k / 52.4B	12.797	.262	.360	.128	.248	.050
Random	fully random 16L	0/16	all	1×10^{-4}	200k / 52.4B	11.498	.247	.360	.111	.209	.063

Table 2: **Multi-axis recovery endpoints.** "Inherited/fresh" counts decoder blocks; copied embeddings, final norm, and head are stated in §3. MMLU-L scores answer letters; MMLU-C scores complete option text by mean log-likelihood per continuation token. PopQA is normalized-answer containment; TriviaQA and NQ-open are exact match. PPL is in-domain on a same-source held-out Dolmino shard. Budgets report optimizer steps and nominal token presentations (effective batch 128, length 2,048). The table exposes the measurement result: similar PPL values can accompany very different target scores, the MMLU verdict depends on interface, and same-depth constructions reach different endpoints. ShortGPT is the strongest compressed endpoint on all three closed-book tasks, but remains far below the intact references. MMLU-L uses the downstream aggregate; the same-item dual-interface rerun is reported separately in Table 5. full32 is reported at 200k. Every trained row is one run.

keep14 checkpoint	128k	153.5k	200k
Held-out PPL↓	10.826	10.693	**10.561**
MMLU letter↑	.3012	.3124	**.3191**
MMLU deficit to base (pp)↓	30.41	29.29	**28.62**

Table 3: **Late keep14 recovery audit.** This table uses the independently merged trajectory/downstream aggregate (.3191 at 200k), not the .3184 dual-interface rerun in Table 5. PPL and MMLU both improve, but the endpoint still has a 28.62-point MMLU deficit to the intact base. On the common 14,042-item rerun, 128k→200k gains 1.68 MMLU points (paired bootstrap 95% CI $[1.08, 2.29]$; exact McNemar $p = 4.12\times10^{-8}$). The interval measures item variation for these fixed checkpoints.

complementary transformations that a contiguous prefix omits. The current contrast changes all four factors together, so it motivates a factorial experiment rather than selecting one explanation. Its immediate design lesson is nonetheless clear: "16 layers" is not a sufficient model specification for recovery.

ShortGPT improves closed-book QA without closing the gap. ShortGPT reaches .159/.330/.067 on PopQA containment, TriviaQA EM, and NQ-open EM, versus .142/.294/.060 for keep14. Its stronger endpoint therefore extends beyond answer letters, yet remains far below the base (.257/.636/.205) and full32 (.228/.572/.158).

Same-shape operating points add a second design lesson. Frozen has worse PPL than Random but higher answer-letter MMLU, whereas both score .360 under the content interface. Across realized checkpoints, keep14 beats Random by 7.14 points and Frozen by 5.60 points on paired MMLU (95% CIs $[6.17, 8.10]$ and $[4.63, 6.59]$). These values are recomputed from the same per-item files used by the interface audit. Initialization, trainable modules, and learning rate must therefore accompany endpoint scores; a shape label and PPL value are not enough to reconstruct the recovery condition.

Table 8 makes the evaluation contrast explicit. The inherited prefix contributes measurable answer-letter signal beyond both same-shape null points, but the content interface largely erases the Frozen–Random distinction. This is exactly why recovery audits need both a paired target comparison and a null-calibrated interface comparison: either view alone would hide part of the result.

6 Additional Scope Checks

6.1 Descriptive variation across MMLU groups

Across the standard broad MMLU groups, keep14 remains below the intact base and ShortGPT remains above keep14. The realized keep14 scores range from .294 on STEM to .377 on Other; ShortGPT improves STEM/humanities/social science/Other by 11.1/12.9/21.9/19.7 points but does not close the base gap. The advantage is therefore broad rather than driven by one group. The complete group and 57-subject inventories remain in Appendix A.4.

6.2 Construction effects beyond MMLU

The complete profile in Table 4 shows that ShortGPT's advantage is not confined to MMLU. Relative to keep14 at displayed step 200k, it gains 4.1 points on HellaSwag, 3.8 on ARC-Challenge, and 9.1 on BoolQ; its core6 macro rises from .5938 to .6215. The gain is therefore broad, but highly

Arm	step	MMLU	HS	ARC-C	ARC-E	PIQA	WinoG	OBQA	LAMB.	BoolQ	CSQA	SIQA
7B base full (32L)	—	**.6053**	.805	.571	.829	.811	.744	.462	.732	.815	.665	.502
keep8 (10L)	200k	**.2535**	.517	.360	.652	.717	.531	.366	.446	.595	.452	.398
keep10 (12L)	200k	**.2718**	.547	.366	.644	.726	.542	.356	.496	.609	.454	.415
keep12 (14L)	200k	**.2752**	.599	.394	.689	.737	.607	.376	.535	.587	.496	.410
keep14 (16L)	200k	**.3191**	.645	.438	.705	.745	.626	.404	.577	.638	.499	.434
ShortGPT-16	200k	**.4739**	.685	.476	.746	.758	.655	.408	.619	.729	.534	.442
frozen-front (16L)	200k	**.2628**	.595	.381	.666	.724	.646	.366	.513	.592	.453	.414
fully random-init (16L)	200k	**.2461**	.578	.414	.697	.733	.545	.384	.484	.614	.450	.416

Table 4: **Complete 11-task checkpoint and endpoint profile.** MMLU here is the independently merged downstream snapshot (.3191 for keep14 at 200k); Table 5 reports a separate paired dual-interface rerun (.3184). `acc_norm` is used for HS, ARC-C/E, PIQA, and OBQA; raw accuracy is used for MMLU, WinoG, LAMBADA, BoolQ, CSQA, and SIQA. Here `acc_norm` means accuracy from *character-normalized* candidate log-likelihood. Appendix Table 17 reports raw and character-normalized sensitivity for the final three tasks. Abbreviations: HS = HellaSwag; ARC-C/E = ARC-Challenge/Easy; WinoG = WinoGrande; OBQA = OpenBookQA; LAMB. = LAMBADA; CSQA = CommonsenseQA; SIQA = SocialIQA. ShortGPT improves every reported task over the keep14 endpoint, but the gain ranges from 0.4 points on OBQA to 15.5 on MMLU. All trained endpoint rows are reported at step 200k; the separate keep14 trajectory appears in Table 3.

Operating point	Letter	Content	Δ (pp)
Intact base	.6054	.4706	−13.48
full32, 200k	.5877	.4662	−12.15
keep14, 200k	.3184	.3832	+6.48
Frozen, 200k	.2624	.3604	+9.80
Random, 200k	.2470	.3598	+11.28
ShortGPT, 200k	.4742	.4012	−7.30

Table 5: **Interface audit on the same 14,042 MMLU items.** This is the dual-interface rerun snapshot; its keep14 letter score is .3184, distinct from the independently merged .3191 trajectory/downstream aggregate. Content uses mean log-likelihood per complete-option token. It raises keep14 by 6.48 points (95% CI $[5.43, 7.56]$), but raises Random by 11.28 points ($[10.24, 12.31]$) to essentially the same content score as Frozen. The random floor shows that an interface gain cannot by itself certify inherited target recovery.

target-dependent: the 15.5-point letter-MMLU difference is much larger than the 2.8-point core6 difference, the 1.8-point content-MMLU difference, and the 1.7/3.6/0.7-point closed-book gains on PopQA/TriviaQA/NQ-open.

This profile strengthens the design lesson from the same-depth comparison. Construction choices do not simply translate every benchmark by a constant amount; they reshape the evaluation profile. Reporting one aggregate endpoint would hide both ShortGPT's broad benefit and the unusually large answer-letter-MMLU effect. Table 4 reports all eleven multiple-choice/likelihood tasks and all available endpoint rows; the closed-book columns are in Table 2.

6.3 The shallow observations are not a depth law

The shallow keep8, keep10, and keep12 arms share the reported 200k-step budget with keep14. This makes the ladder a matched-step endpoint comparison, while unequal model depth still implies unequal FLOPs. It therefore supports a depth-conditioned endpoint contrast, not a compute-normalized ordering or a universal architectural threshold.

Within keep8, several likelihood-style tasks and PPL improve over the observed interval while aggregate MMLU remains near chance. Together with the keep14 trajectory, this identifies matched stopping-rule and compute-normalized depth experiments as the next step.

6.4 Scope checks do not constitute replications

A more compressed OLMo-2-1B prefix+fresh-tail run shows falling PPL with near-chance MMLU, and an existing Qwen3-8B endpoint is also near chance after 200k. The 1B arm changes scale and retained fraction; the Qwen arm additionally changes architecture, corpus, and available evaluations. We include them as directional context only, not as matched replications or evidence of cross-family universality (Appendix Table 14).

6.5 Synthesis

Across these scope checks, the same audit logic holds: a target score must be interpreted together with its interface, construction, and training budget. The main OLMo-2-7B study supports this conclusion directly; the shallow, 1B, and Qwen observations indicate where matched replications

PPL corpus	Base	full32	ShortGPT	keep14	Random	Frozen
Dolmino held-out	7.40	7.67	9.78	10.56	11.50	12.80
WikiText-103	6.00	6.77	10.58	11.56	13.45	13.66
PG-19	10.18	10.76	14.45	15.43	17.54	17.35

Table 6: **The likelihood ordering transfers out of domain.** Token-weighted PPL uses the same no-chat, no-BOS evaluator on 289k WikiText-103 tokens and 2.46M PG-19 tokens. The ordering Base≈full32 < ShortGPT < keep14 < Random/Frozen is unchanged. Thus the same-source PPL result is not hiding an OOD reversal; the compressed models pay an equal or larger relative likelihood tax outside Dolmino.

Construction	Inherited	Fresh	PPL↓	MMLU-L	MMLU-C
keep14+fresh2	0–13	2	10.561	.319	.383
ShortGPT-16	0–12,16,17,31	0	**9.780**	**.474**	**.401**
ShortGPT gain	same 16L / 200k		−0.781	+15.5 pp	+1.8 pp

Table 7: **What the same-depth contrast reveals.** Both models have 16 blocks and are displayed at step 200k with the same peak LR; ShortGPT is 15.5 letter-MMLU points and 1.8 content-MMLU points stronger. It jointly retains two more pretrained blocks, uses non-contiguous selection, preserves block 31, and removes the fresh tail. The contrast identifies construction as a first-class recovery variable and motivates a factorial follow-up over these four choices.

MMLU letter comparison	Δ (pp)	95% CI	McNemar p
keep14 − Random	+7.14	$[6.17, 8.10]$	9.19×10^{-47}
keep14 − Frozen	+5.60	$[4.63, 6.59]$	8.08×10^{-28}
Frozen − Random	+1.54	$[0.48, 2.58]$	4.00×10^{-3}

Table 8: **Paired same-shape operating-point differences.** All rows are recomputed from the same six-arm anonymous-artifact snapshot on 14,042 items at displayed step 200k; intervals use 10,000 paired bootstrap resamples (seed 0) and exact two-sided McNemar tests. Inherited keep14 retains a clear letter-interface advantage over both null points even though Frozen and Random converge to the same .360 content score (Table 5).

would be most useful.

7 Discussion

Recovery is a vector, not a loss threshold. Perplexity remains the correct measure of the CPT objective, and the keep14 trajectory shows that continued optimization also produces a real MMLU gain. What it does not provide is a certificate that a target behavior has returned. The intact-base gap, OOD ordering, interface audit, closed-book results, and low continuation-stream overlap show why recovery should be represented by a vector rather than the direction of one loss curve.

Null baselines turn protocol sensitivity into evidence. Without Random, the 6.48-point keep14 content gain could be read as revealing capability hidden by answer letters. The 11.28-point Random gain instead shows that much of this interface is attainable as option-text compatibility after training from scratch. This is a general evaluation principle: when a new interface is introduced to reveal latent performance, its null floor is part of the construct-validity test.

A practical recovery audit. We recommend reporting six axes for post-intervention recovery: (1) in-domain and out-of-domain likelihood; (2) target evaluations with sample counts; (3) prompt/scoring/generation interface and a relevant null baseline; (4) exact inherited/fresh structure and trainable modules; (5) steps, token presentations, compute, and stopping rule; and (6) training-run replication separately from item-level uncertainty. These axes make recovery claims comparable even when pruning algorithms, training budgets, and evaluation interfaces differ.

8 Conclusion

Our OLMo-2-7B audit shows that keep14 improves yet retains a 28.6-point MMLU deficit; its content-interface gain is smaller than the random-null gain; and the gap recurs in closed-book QA. At the same depth and displayed steps, ShortGPT is 15.5 letter-MMLU points stronger and improves all three closed-book tasks, while coupling four construction factors. The likelihood ordering also transfers to WikiText-103 and PG-19, and measured continuation-stream overlap does not explain the gaps. Recovery claims should therefore report likelihood, target behavior, interface, construction, budget, and uncertainty together.

Limitations

The principal keep14 path, ShortGPT, and the same-shape operating points are single training runs. Item-level bootstrap intervals and McNemar tests condition on fixed checkpoints; they do not measure variation from initialization, data order, optimization, or block selection. Training seeds were not explicitly set in the historical runs.

full32 and keep14 are both reported at 200k, but each construction remains a single run and equal optimizer steps do not imply equal FLOPs. The shallow rows share the reported 200k-step and nominal-token budget with keep14, but the study does not match FLOPs across depths. Random-init changes LR and all initialization; Frozen changes the trainable set; ShortGPT changes four construction factors. These comparisons bound alternatives but do not identify causes.

Letter and content MMLU simultaneously change prompt, candidate text, tokenization, and normalization. Paired interface artifacts exist for the 14,042 MMLU items, but the comparison remains multi-factor. Closed-book evaluations report complete split sizes and exact normalization; ShortGPT is now evaluated on all three splits. Prompt-free per-item ShortGPT score records are released, but aligned records for every other closed-book arm were not consolidated, so we do not report paired closed-book intervals. OOD PPL covers WikiText-103 and PG-19, not a broad domain suite. The contamination scan is token-overlap evidence, not a guarantee that no semantic paraphrase or pretraining exposure exists; many short PopQA/NQ questions are undecidable under 13-grams, although their 8-gram flagged rates remain below 1%.

The study is English, mainly 7B, one model family, one mixture, and one prefix+fresh-tail recipe. It does not report latency, throughput, memory, or recovery FLOPs. Per-run and project-wide GPU-hours are unavailable because historical hardware/wall-time records were incomplete. Exact reproduction of keep14 is further blocked by an unrecorded within-epoch loader offset after a 34.5k resume. These omissions cannot be repaired by writing alone.

The anonymous artifact snapshot releases the available evaluator scripts, sanitized configs/manifests, aggregate summaries, prompt-free per-item MMLU artifacts, and paired-analysis outputs. The original evaluator revisions were local-only commits; the snapshot preserves their file contents, not public commit ancestry. It also cannot recover the historical training seed or lost loader offset. No layer-wise readout is used to explain the behavioral observations.

Ethical Considerations

This work adds no safety mechanism and inherits risks of the underlying model and data, including hallucination, bias, and unsafe generation. Its practical caution is measurement-related: better in-domain perplexity need not imply recovery of capabilities required in deployment, so target evaluations and relevant null baselines should be checked directly.

We collect no human-subject data or annotations. Experiments use released models, corpora, and benchmarks under their upstream terms. Training and large-scale evaluation consume energy; incomplete historical records prevent a reliable GPU-hour total. Released artifacts should preserve upstream licenses and exclude benchmark text or model weights that cannot be redistributed.

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A Complete Results

The main paper emphasizes an observed-budget, single-run recovery gap. This appendix reports all consolidated checkpoints, including the qualitative 1B comparison, within-arm trajectories, and chance-adjusted summaries.

A.1 Held-out perplexity checkpoints

Table 9 distinguishes progressing checkpoints from true endpoints. The fully random-init arm is a random language model with the same 16-layer architecture: its decoder blocks, embedding, final norm, and output head are initialized without OLMo-2 weights. It uses a different learning rate, so it is a budgeted operating point rather than an initialization-only ablation. Table 10 reports the step-zero and 200k non-contiguous operating points. Its 16 retained layers are selected by cosine block influence on 128 Dolmino windows and include the original final layer.

A.2 Within-architecture training trajectories

The complete 11-task keep8 trajectory in Table 11 gives a within-arm view over recorded checkpoints. From 44k to the 200k endpoint, most reported tasks improve, while aggregate MMLU changes from .2463 to .2535 and remains near chance. Because no common per-item 44k prediction set or replicated training run is available, we use this only as descriptive evidence against a uniform late gain over the observed interval, not against recovery after more training.

The OLMo-2-0425-1B arm shows the same qualitative separation (Figure 2; Tables 12–13): PPL falls $17.619 \rightarrow 15.628$ and HellaSwag/ARC-Easy/PIQA improve, while MMLU is $.2495/.2558/.2529/.2480$. Because this is one more-compressed arm from the same model family with a weaker base, we treat it as qualitative context, not an independent replication.

A.3 Target evaluations and continued pretraining

Main-text Table 4 contains the complete 11-task profile; Table 2 provides the compact multi-axis endpoint summary. Table 15 exposes why a single average benchmark score would be misleading. Fully random initialization nearly matches inherited keep14 on ARC-Easy, PIQA, and BoolQ, but inherited initialization retains more signal on WinoGrande, MMLU, and several other downstream

Scale	Arm	step	inherited	total layers	PPL↓	tax↓
7B	full base	—	32	32	7.398	1.000×
	keep8+fresh2	5k	8	10	22.331	3.019×
	keep8+fresh2	15k	8	10	17.868	2.416×
	keep8+fresh2	25k	8	10	16.426	2.220×
	keep8+fresh2	35k	8	10	15.612	2.110×
	keep8+fresh2	44k	8	10	15.131	2.045×
	keep8+fresh2	200k†	8	10	13.333	1.802×
	keep10+fresh2	10k	10	12	17.239	2.330×
	keep10+fresh2	200k†	10	12	12.816	1.732×
	keep12+fresh2	111.5k	12	14	11.566	1.563×
	keep12+fresh2	200k†	12	14	11.443	1.547×
	keep14+fresh2	128k	14	16	10.826	1.463×
	keep14+fresh2	153.5k	14	16	10.693	1.445×
	keep14+fresh2	200k	14	16	10.561	1.428×
	full32 continued@200k	200k‡	32	32	7.670	1.037×
	ShortGPT-16	200k	16	16	**9.780**	**1.322×**
	frozen-front control	200k	14	16	12.797	1.730×
	fully random-init control	200k	0	16	11.498	1.554×
1B	full base	—	16	16	10.642	1.000×
	keep7+fresh2	50k	7	9	17.619	1.656×
	keep7+fresh2	100k	7	9	16.161	1.519×
	keep7+fresh2	200k	7	9	15.628	1.469×

Table 9: **All held-out perplexity checkpoints used in this study.** Every point uses the same disjoint 4096×2048 Dolmino validation shard (8,384,512 target tokens) and token-weighted eight-shard merging. The frozen-front arm retains the same pretrained prefix but does not update it; the fully random-init arm has the same 16-layer shape but loads no pretrained weights. ShortGPT inherits non-contiguous layers $[0–12, 16, 17, 31]$ and appends no fresh blocks. †The keep8, keep10, and keep12 endpoint rows use the unified 200k endpoint convention. Earlier measurements retain their true intermediate checkpoint labels; only the final endpoint row is normalized to 200k. ‡The full32 values are reported under the unified 200k endpoint convention.

tasks. The late keep14 audit is reported in the main paper (Table 3), while Table 15 retains the broader endpoint and recovery summary. Table 17 additionally exposes raw and character-normalized scoring for BoolQ, CommonsenseQA, and SocialIQA, whose options have unequal lengths; the main tables consistently use raw accuracy for these tasks. The paired MMLU interface audit and headline closed-book results now appear in the main paper (Tables 5 and 2); exact protocols and sample counts appear below. The paired same-shape MMLU comparisons have moved to the main paper (Table 8); their intervals quantify evaluation-item variation rather than training-seed uncertainty.

The confidence intervals in Table 16 distinguish the available checkpoints conditionally: keep8 and fully random init include chance, while keep10, keep12, and keep14 are above it. These are not between-training-run intervals. The subject-group table below describes heterogeneity rather than testing domain-specific effects.

ShortGPT point	PPL	core6	MMLU
step 0	401.124	—	.2620
step 200k	**9.780**	**.6215**	**.4739**

Table 10: **Non-contiguous ShortGPT endpoint.** For block i, selection uses $\mathrm{BI}_i = 1 - \mathbb{E}_{x,t} \cos(h_{\mathrm{in}}^{(i)}, h_{\mathrm{out}}^{(i)})$ over 128 evenly spaced Dolmino windows (262,144 tokens), retaining the 16 highest-BI blocks $[0\text{–}12, 16, 17, 31]$. No evaluation task is used for selection and no fresh tail is appended. `core6` is the unweighted macro-average of HellaSwag, ARC-Challenge, ARC-Easy, PIQA, and OpenBookQA character-normalized accuracy plus WinoGrande raw accuracy. The unhealed construction is initially degenerate; the 200k endpoint is one run and is not a selection-only ablation.

Task	metric	10k	25k	44k	200k	$\Delta_{44\to 200k}$
MMLU	raw	.2542	.2502	.2463	.2535	$+0.72$
HellaSwag	norm.	.3915	.4390	.4694	.5169	$+4.75$
ARC-Challenge	norm.	.2654	.3114	.3140	.3601	$+4.61$
ARC-Easy	norm.	.5581	.5930	.6103	.6519	$+4.16$
PIQA	norm.	.6632	.6801	.6937	.7165	$+2.28$
WinoGrande	raw	.5209	.5083	.5185	.5312	$+1.27$
OpenBookQA	norm.	.3000	.3360	.3320	.3660	$+3.40$
LAMBADA	raw	.3429	.3827	.4333	.4463	$+1.31$
BoolQ	raw	.5713	.6080	.5881	.5948	$+0.67$
CommonsenseQA	raw	.3923	.4103	.4423	.4521	$+0.98$
SocialIQA	raw	.3767	.3874	.4002	.3976	-0.26

Table 11: **Complete 11-task keep8 downstream trajectory.** The final column is the accuracy-point change from 44k to the reported 200k endpoint under the reported metric shown in column two; "norm." denotes accuracy based on character-normalized candidate log-likelihood. MMLU remains near chance while most other tasks improve. Earlier PPL checkpoints use a separate grid; the endpoint held-out PPL is reported at 200k in Table 9. This table reports aggregate checkpoints, not a paired 44k–200k test or a replicated trajectory.

A.4 MMLU subject-group results

Table 18 quantifies the broad-domain pattern. The inherited keep14 model is strongest in social science and the heterogeneous "Other" group, whereas science, technology, engineering, and mathematics (STEM) and humanities remain closer to chance. Table 20, placed at the end of the appendix as two side-by-side panels, reports every one of the 57 subjects at the 153.5k and final 200k keep14 checkpoints.

B Evaluation and Reproducibility Details

B.1 Training recipe and unavailable fields

Training uses English OLMo-2/Dolmino artifacts. The DCLM portion of `dolmino-mix-1124` is packed into 2,048-token windows; the training array contains 7,570,911 windows and the fixed in-domain validation array contains 4,096 disjoint windows. Exact SHA-256 values for the training array, validation array, and five headline checkpoints are listed in the accompanying anonymous checksum manifest; the validation array begins `ee36248dc8e4a79c` and ends `2f0735e570f22e8`. All headline arms use effective batch size 128, 150 warmup steps, cosine decay, AdamW (β_1=0.9, β_2=0.95, ϵ=10^{-8}), matrix weight decay 0.1, vector weight decay 0, and gradient clipping at 1.0. Parameters and optimizer state are fp32; forward passes use bf16 autocast; gradient checkpointing is enabled.

The executed keep14 and frozen-front runs assigned every trainable tensor to the $2 \times 10^{-5}/2 \times 10^{-6}$ peak/minimum LR group; random-init used $10^{-4}/10^{-5}$; ShortGPT and full32 used $2 \times 10^{-5}/2 \times 10^{-6}$. Exact trainable sets and budgets appear in Table 2. Launch-time reconstruction checks require exact copied-tensor equality ($\max |\theta_{\mathrm{arm}} - \theta_{\mathrm{base}}| = 0$), and evaluation rebuilds the architecture from checkpoint metadata before a strict state-dictionary load.

Historical training seeds are unavailable. keep14 resumed at 34.5k with model, optimizer, and RNG state restored, but the checkpoint omitted the within-epoch loader offset; the resumed iterator

Model	step	HS	ARC-C	ARC-E	PIQA	WinoG	OBQA	PPL
1B full base	—	.683	.422	.735	.757	.643	.400	10.642
keep7+fresh2	50k	.411	.289	.568	.671	.531	.318	17.619
keep7+fresh2	100k	.438	.300	.593	.687	.526	.326	16.161
keep7+fresh2	147k	.450	.311	.595	.692	.530	.326	15.628
keep7+fresh2	200k	.448	.294	.606	.695	.544	.330	—

Table 12: **OLMo-2-1B core-task trajectory.** We report `acc_norm` (character-normalized candidate-likelihood accuracy) except for WinoGrande, where raw and normalized accuracy coincide. Abbreviations: HS = HellaSwag; ARC-C/E = ARC-Challenge/Easy; WinoG = WinoGrande; OBQA = OpenBookQA; PPL = perplexity.

Model	step	MMLU	LAMBADA	BoolQ	CSQA	SIQA
1B full base	—	.3820	.6480	.624	.553	.422
keep7+fresh2	50k	.2495	.3536	.618	.368	.384
keep7+fresh2	100k	.2558	.3740	.597	.380	.394
keep7+fresh2	147k	.2529	.4021	.584	.391	.384
keep7+fresh2	200k	.2480	.3969	.577	.387	.386

Table 13: **OLMo-2-1B MMLU and likelihood-task trajectory.** MMLU remains at the .25 chance floor across the complete measured trajectory, whereas LAMBADA and commonsense tasks improve. Raw accuracy is shown for these five tasks.

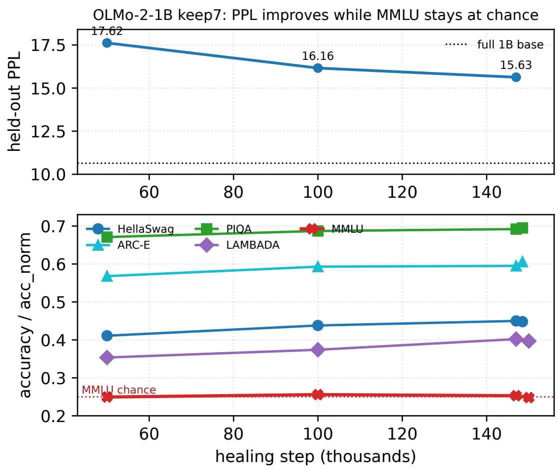

Figure 2: **Qualitative OLMo-2-1B trajectory.** PPL and several ordinary tasks improve while MMLU stays near chance over the observed keep7 interval.

Qwen3-8B arm	Depth	PPL↓	MMLU letter
Full base	36/36	11.42	.7297
keep12 + fresh2, 200k	14/36	23.49	.2495

Table 14: **Cross-family endpoint diagnostic.** The Qwen3-8B (Yang et al., 2025) arm heals on SlimPajama rather than Dolmino and is more compressed than the principal OLMo arm. It supports only the directional answer-letter finding; absolute PPL and recovery magnitude are not compared across families. This endpoint does not include the content-MMLU and closed-book diagnostics used in the principal study.

restarted that epoch's distributed shuffle. Per-run wall time/GPU-hours and aggregate project compute are unavailable. The currently retained arrays and headline checkpoints have SHA-256 checksums, but no historical checksum manifest was recorded at execution time. We report nominal token presentations rather than unique-token counts and do not infer missing values.

B.2 Evaluation protocols and sample counts

Perplexity is merged as $\exp(\sum_g \mathrm{NLL}_g / \sum_g n_g)$ over eight shards; per-shard perplexities are never averaged. At keep14 128k, 153.5k, and 200k, independent merging reproduces the stored PPL and downstream aggregates to 10^{-9}.

For MMLU letter scoring, the prompt contains the subject description, question, all four options labeled A–D, and `Answer:`; the candidates are the four answer letters. For content scoring, the prompt retains the subject description and question but removes the lettered option body; the candidates are the four complete option texts. Both protocols use the same 14,042 item IDs, tokenizer, no-BOS rule, and truncation. Each saved per-item record contains option-level summed scores; content records additionally contain continuation token counts and normalized scores. The anonymous artifact directory contains the six headline-arm per-item content-interface files, summaries, and paired trajectory outputs, without benchmark text.

Closed-book evaluation is zero-shot and no-retrieval. The exact prompt is `Question: {q}\nAnswer:`; decoding is greedy with `do_sample=False`, at most 32 new tokens, no chat template, and no inserted BOS. The first completion line is lowercased, stripped of punctuation and articles, and whitespace-normalized. Scores are maximized over all aliases. PopQA uses test (n=14,267) and containment; TriviaQA uses `rc.nocontext` validation (n=17,944) and exact match; NQ-open uses validation (n=3,610) and exact match. EM, containment, and token F1 were retained. The artifact contains all three prompt-free ShortGPT per-item score files and aggregate summaries for every headline arm. Aligned per-item records for the other arms were not all consolidated, so the manuscript does not report paired closed-book intervals.

B.3 OOD likelihood and continuation-stream overlap

The OOD arrays contain 288,627 WikiText-103 and 2,456,400 PG-19 target tokens. All six endpoint models use the same raw next-token evaluator, tokenizer, no-chat/no-BOS interface, and token-weighted aggregation as the Dolmino held-out result. The exact values appear in Table 6; the released summary records corpus token counts, sum-NLL aggregation, and each relative-to-base ratio.

For overlap, we tokenize every benchmark item with the OLMo tokenizer and scan the exact 15,505,225,728-token continuation stream once with an inverted index. An item is flagged when at least 80% of its token n-grams occur in the stream; both $n = 13$ and $n = 8$ are reported. Short items without a complete 13-gram are marked undecidable rather than clean. The MMLU sensitivity recompute removes flagged items from the saved keep14 and Random per-item predictions; the gap changes by at most 0.17 points. This audit detects near-verbatim token overlap, not semantic paraphrases or exposure in the original OLMo pretrain-

Family	Task	chance	base	inherited	random init	recovery (%)
				reported score		*inherited / random*
surface	ARC-Easy	.25	.829	.705	.697	78.6/77.2
surface	PIQA	.50	.811	.745	.733	78.9/74.9
reasoning	HellaSwag	.25	.805	.645	.578	71.1/59.1
reasoning	ARC-Challenge	.25	.571	.438	.414	58.5/51.1
reasoning	WinoGrande	.50	.744	.626	.545	51.6/18.4
reasoning	OpenBookQA	.25	.462	.404	.384	72.6/63.2
reasoning	LAMBADA	.00	.732	.577	.484	78.9/66.1
reasoning	SocialIQA	.333	.502	.434	.416	59.8/49.1
reasoning	CommonsenseQA	.20	.665	.499	.450	64.3/53.9
in-context	BoolQ	.50	.815	.638	.614	43.9/36.2
answer-letter	**MMLU**	.25	.6053	.3191	.2461	$\mathbf{19.4}/-\mathbf{1.1}$

Table 15: **Exact above-chance recovery for inherited keep14 and the fully random-init control at the matched step 200k.** Recovery is $100(x-c)/(b-c)$ for score x, chance c, and full-base score b. Family labels are descriptive: "surface" contains ARC-Easy and PIQA; "in-context" denotes passage-grounded BoolQ; "answer-letter" denotes the MMLU protocol; the remaining tasks are grouped descriptively as "reasoning." Reported scores use character-normalized candidate likelihood for HellaSwag, ARC, PIQA, and OpenBookQA and raw accuracy otherwise. The negative random-init MMLU value means statistically chance-level performance. The random-init arm also randomizes lexical modules and uses a different learning rate, so the table is an operating-point comparison rather than an initialization-only or decoder-block-only ablation. Negative recovery denotes a score below the chance-adjusted reference.

Arm	MMLU	SE	approx. 95% CI
full 7B base	.6053	.0041	[.5972, .6134]
keep8, 200k	.2535	.0037	[.2463, .2607]
keep10, 200k	.2718	.0038	[.2644, .2792]
keep12, 200k	.2752	.0038	[.2678, .2826]
keep14, 128k	.3012	.0039	[.2936, .3088]
keep14, 153.5k	.3124	.0039	[.3047, .3201]
keep14, 200k	.3191	.0039	[.3114, .3268]
ShortGPT-16, 200k	.4739	.0042	[.4656, .4822]
frozen-front, 200k	.2628	.0037	[.2555, .2701]
fully random init, 200k	.2461	.0036	[.2390, .2532]

Table 16: **Marginal MMLU uncertainty across the 7B frontier.** Standard errors and Wald intervals use $n = 14{,}042$ questions. keep8 and fully random init include the .25 chance floor; keep10/keep12, frozen-front, all keep14 checkpoints, and ShortGPT are above it. These marginal intervals summarize each arm separately; Table 8 gives the stronger item-paired three-control comparisons.

Task	metric	base	k8	k12	k14-153	k14-200	frozen	random
BoolQ	raw	.815	.588	.610	.606	.638	.592	.614
BoolQ	norm.	.753	.620	.657	.682	.689	.700	.619
CSQA	raw	.665	.442	.508	.506	.499	.453	.450
CSQA	norm.	.652	.405	.471	.479	.476	.467	.405
SIQA	raw	.502	.400	.415	.441	.434	.414	.416
SIQA	norm.	.547	.431	.460	.474	.474	.451	.443

Table 17: **Raw versus character-normalized accuracy.** The normalized score divides candidate log-likelihood by the raw continuation's character count before selecting an answer. This is distinct from the token-normalized complete-option MMLU protocol in Table 5. The main results report raw accuracy for BoolQ, CommonsenseQA, and SocialIQA. Normalization can materially change BoolQ and SIQA, so both views are reported here. k14-153/k14-200 denote the true 153.5k and 200k keep14 checkpoints; frozen-front and random init are 200k operating points.

ing corpus.

We also recompute the three closed-book tasks on the strict 13-gram clean subset. TriviaQA removes 142 of 17,944 examples and NQ-open removes one of 3,610; PopQA has zero flagged items and is unchanged. Across the seven available arms, the largest absolute headline-metric shift is 0.15 points and the ordering Base > full32 > ShortGPT > keep14 > Frozen/keep8/Random is preserved. For keep14, TriviaQA changes 29.40→29.27 and NQ-open 5.98→5.99; for ShortGPT the corresponding changes are 33.01→32.87 and 6.68→6.68.

B.4 Evaluator and prediction provenance

The source package includes `anonymous_artifact/` with evaluator scripts, protocol/config manifests, SHA-256 checksums, aggregate closed-book summaries, prompt-free ShortGPT closed-book per-item score files, six headline-arm content-MMLU per-item score files, endpoint OOD-PPL/contamination summaries, and the keep14 128k/200k paired trajectory artifacts. The local-only evaluator revisions are recorded in a manifest and their available source files are snapshotted; we do not

MMLU group	n	Base	full32	keep14	ShortGPT
STEM	3,864	.534	.507	.294	.405
Humanities	4,705	.555	.542	.299	.428
Social science	3,077	.709	.690	.335	.554
Other	2,396	.686	.670	.377	.574

Table 18: **The ShortGPT advantage spans all broad MMLU groups.** ShortGPT exceeds keep14 by 11.1, 12.9, 21.9, and 19.7 points, respectively. The largest differences occur in social science and Other rather than one isolated subject family. full32 is the reported 200k intact-CPT endpoint; keep14 and ShortGPT are 200k endpoints.

Task	split / mode	n	chance	metric
HellaSwag	validation, 4-choice	10,042	.25	acc_norm
ARC-Challenge	test, 4-choice	1,172	.25	acc_norm
ARC-Easy	test, 4-choice	2,376	.25	acc_norm
PIQA	validation, 2-choice	1,838	.50	acc_norm
WinoGrande	validation, 2-choice cloze	1,267	.50	accuracy
OpenBookQA	test, 4-choice	500	.25	acc_norm
MMLU	test, 57 subjects, 4-choice letters	14,042	.25	accuracy
LAMBADA-OpenAI	test, greedy final word	5,153	≈ 0	exact greedy accuracy
BoolQ	validation, yes/no	3,270	.50	accuracy
CommonsenseQA	validation, 5-choice	1,221	.20	accuracy
SocialIQA	validation, 3-choice	1,954	.333	accuracy
PopQA	test, open generation	14,267	—	containment
TriviaQA	rc.nocontext validation	17,944	—	exact match
NQ-open	validation	3,610	—	exact match

Table 19: **Evaluation suite, sample counts, and headline metrics.** The first eleven rows use zero-shot likelihood scoring, except LAMBADA's greedy final-word continuation. For HellaSwag, ARC, PIQA, and OpenBookQA, `acc_norm` divides summed candidate log-likelihood by the raw continuation's character count. The complete-option MMLU control instead uses continuation-token normalization. The last three rows use greedy, zero-shot, no-retrieval generation with the fixed `Question:/Answer:` prompt. No chat template or inserted BOS is used. All listed examples are evaluated.

claim public commit ancestry. The snapshot excludes benchmark text, private paths, credentials, model weights, and unconsolidated closed-book generations. It also does not infer the unavailable historical seed or loader offset.

For artifact auditing, the base content-MMLU per-item JSONL has 14,042 rows; its SHA-256 begins `52f60373c952dd86` and ends `c1e07766e0364701`. Each row records item ID, subject, gold answer, raw option scores for letter/content, normalized content scores, continuation token counts, and correctness. The within-keep14 128k/200k trajectory analysis uses 10,000 bootstrap resamples with seed 1234. The same-shape endpoint comparisons in Table 8 instead use the six-arm `mmlu_content` snapshot, 10,000 resamples with seed 0, and exact two-sided McNemar tests. Both are item-level procedures conditional on fixed checkpoints, but they are not mixed into one comparison.

Subject	n	base	k8	k12	k14-153	k14-200	frozen	rand
abstract_algebra	100	.310	.300	.240	.260	.270	.310	.240
anatomy	135	.622	.289	.296	.363	.363	.296	.215
astronomy	152	.730	.171	.289	.316	.322	.283	.178
business_ethics	100	.590	.250	.340	.390	.430	.260	.220
clinical_knowledge	265	.649	.242	.275	.317	.317	.257	.249
college_biology	144	.771	.278	.271	.347	.326	.250	.264
college_chemistry	100	.470	.190	.310	.270	.230	.260	.200
college_computer_science	100	.490	.240	.300	.300	.290	.240	.270
college_mathematics	100	.360	.200	.310	.240	.320	.270	.230
college_medicine	173	.566	.277	.341	.295	.289	.249	.214
college_physics	102	.422	.196	.304	.225	.275	.265	.147
computer_security	100	.680	.230	.250	.410	.400	.340	.290
conceptual_physics	235	.570	.238	.264	.311	.332	.243	.277
econometrics	114	.360	.246	.193	.246	.246	.325	.237
electrical_engineering	145	.600	.255	.276	.359	.366	.345	.248
elementary_mathematics	378	.397	.278	.254	.296	.272	.259	.222
formal_logic	126	.389	.302	.325	.198	.262	.183	.302
global_facts	100	.360	.340	.330	.280	.290	.320	.190
high_school_biology	310	.765	.223	.319	.319	.319	.239	.190
high_school_chemistry	203	.493	.261	.246	.256	.232	.202	.167
high_school_computer_science	100	.590	.290	.220	.300	.290	.280	.290
high_school_european_history	165	.752	.236	.236	.242	.309	.194	.206
high_school_geography	198	.783	.242	.308	.384	.328	.288	.207
high_school_government_and_politics	193	.870	.202	.285	.368	.358	.244	.197
high_school_macroeconomics	390	.636	.226	.338	.295	.321	.285	.221
high_school_mathematics	270	.296	.230	.278	.281	.248	.259	.219
high_school_microeconomics	238	.655	.239	.269	.277	.277	.252	.185
high_school_physics	151	.404	.238	.358	.225	.245	.298	.212
high_school_psychology	545	.807	.196	.297	.317	.358	.229	.207

Subject	n	base	k8	k12	k14-153	k14-200	frozen	rand
high_school_statistics	216	.514	.171	.338	.204	.213	.218	.194
high_school_us_history	204	.824	.314	.230	.319	.304	.299	.309
high_school_world_history	237	.797	.253	.270	.325	.354	.274	.253
human_aging	223	.650	.283	.139	.368	.359	.265	.341
human_sexuality	131	.756	.221	.298	.382	.382	.298	.321
international_law	121	.777	.248	.240	.281	.264	.355	.240
jurisprudence	108	.685	.204	.269	.287	.315	.231	.259
logical_fallacies	163	.693	.233	.215	.380	.350	.209	.221
machine_learning	112	.455	.170	.277	.375	.330	.277	.321
management	103	.777	.204	.359	.320	.369	.282	.194
marketing	234	.825	.286	.274	.440	.440	.286	.312
medical_genetics	100	.740	.170	.300	.370	.400	.340	.320
miscellaneous	783	.803	.271	.278	.436	.457	.268	.244
moral_disputes	346	.694	.225	.260	.329	.321	.272	.237
moral_scenarios	895	.279	.238	.238	.244	.253	.244	.238
nutrition	306	.693	.239	.307	.324	.327	.320	.242
philosophy	311	.711	.235	.270	.322	.318	.257	.222
prehistory	324	.741	.318	.275	.346	.358	.299	.238
professional_accounting	282	.415	.266	.245	.238	.241	.262	.280
professional_law	1534	.460	.244	.246	.259	.272	.245	.263
professional_medicine	272	.570	.199	.239	.228	.232	.301	.301
professional_psychology	612	.611	.243	.247	.320	.302	.260	.275
public_relations	110	.627	.282	.236	.291	.327	.309	.236
security_studies	245	.727	.367	.339	.294	.335	.245	.245
sociology	201	.836	.289	.279	.418	.428	.194	.294
us_foreign_policy	100	.850	.230	.340	.480	.450	.270	.340
virology	166	.524	.241	.313	.355	.386	.283	.313
world_religions	171	.836	.246	.292	.491	.509	.287	.322

Table 20: **Complete 57-subject MMLU results.** All values are accuracy. k14-153 and k14-200 are the true 153.5k and 200k keep14 checkpoints; frozen-front and random init are the two other 200k same-shape operating points. Subjects are split alphabetically across the two panels.